

MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 5 Solutions

Ex. 3.4.4. Use universal instantiation or universal modus ponens to fill in a valid conclusion for the following argument.

$\forall$ real number r , a , and b , if r is positive, then $(r^a)^b = r^{ab}$.

$r = 3$, $a = 1/2$, and $b = 6$ are particular real numbers such that r is positive.

$\therefore$ _____

SOLUTION . $\therefore (3^{1/2})^6 = 3^{(\frac{1}{2})(6)}$.

□

Ex. 3.4.17. The argument below could be valid by universal modus ponens or universal modus tollens, or invalid and exhibits the converse or the inverse error. State whether it is valid or invalid. Justify your answer.

If an infinite series converges, then the terms go to 0.

The terms of the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ go to 0.

$\therefore$ the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ converges.

SOLUTION . Invalid. This argument exhibits the converse error.

□

Ex. 4.1.13. Disprove the statement by giving a counterexample.

For all integers m and n , if $2m + n$ is odd then m and n are both odd.

SOLUTION . Let $m = 2$ and $n = 1$. Then $2m + n = 2 \cdot 2 + 1 = 5$, which is odd. But m is not odd, and so it is false that both m and n are odd.

□

Ex. 4.1.40. Find the mistake(s) in the “proof” shown below.

Theorem: For all integers k , if $k > 0$ then $k^2 + 2k + 1$ is composite.

“Proof: Suppose k is any integer such that $k > 0$. If $k^2 + 2k + 1$ is composite, then $k^2 + 2k + 1 = r \cdot s$ for some integers r and s such that

$$1 < r < (k^2 + 2k + 1)$$

and

$$1 < s < (k^2 + 2k + 1).$$

Since

$$k^2 + 2k + 1 = r \cdot s$$

and both r and s are strictly between 1 and $k^2 + 2k + 1$, then $k^2 + 2k + 1$ is not prime. Hence $k^2 + 2k + 1$ is composite as was to be shown.”

SOLUTION . The incorrect proof is begging the question. There is no justification to why the integers r and s exists, or rather, the justification is assuming (instead of proving) that $k^2 + 2k + 1$ is composite. $\square$

Ex. 4.1.56. Determine whether the statement is true or false. Justify your answer with a proof or a counterexample, as appropriate.

(Two integers are **consecutive** if and only if one is one more than the other.) Any product of four consecutive integers is one less than a perfect square.

SOLUTION . Consider any product of four consecutive integers. Call the smallest of the four integers n , and the product of the four integers N . Then

$$N = n \cdot (n+1) \cdot (n+2) \cdot (n+3) = [n \cdot (n+3)] [(n+1)(n+2)] = [n^2 + 3n] [n^2 + 3n + 2],$$

which has the form $M \cdot (M+2)$, where $M = n^2 + 3n$. Completing the square, $M \cdot (M+2) = (M+1)^2 - 1$. We apply that on N to get

$$N = [n^2 + 3n + 1]^2 - 1.$$

Clearly $n^2 + 3n + 1$ is an integer because sums and products of integers are integers.

Hence N is one less than the square of $n^2 + 3n + 1$.

Hence the given product of four consecutive integers is one less than a perfect square. $\square$

Ex. 4.2.8. The zero property says that if a product of two real numbers is 0, then one of the numbers must be 0.

- Write this property formally using quantifiers and variables.
- Write the contrapositive of your answer to part (a).
- Write an informal version (without quantifier symbols or variables) for your answer to part (b).

SOLUTION . a. $\forall$ real numbers x and y , if $xy = 0$ then $x = 0$ or $y = 0$.

b. $\forall$ real numbers x and y , if $x \neq 0$ and $y \neq 0$ then $xy \neq 0$.

c. If two real numbers are both nonzero, then the product is nonzero. $\square$

Ex. 4.2.16. Determine if the statement is true or false. If it is true, prove it directly from the definitions, and if it is false, give a counterexample. In case the statement is false, determine whether a small change would make it true. If so, make the change and prove the new statement.

The quotient of any two rational numbers is a rational number.

SOLUTION . Counterexample: Both 1 and 0 are rational numbers but $1/0$ is not a rational number (because it is not even a number since division by 0 is not defined).

Modified Statement: The quotient of any rational number and any nonzero rational number is rational.

Proof of Modified Statement: Suppose r and s are [particular but arbitrarily chosen] rational numbers with $s \neq 0$. By definition of rational,

$$r = \frac{a}{b} \quad \text{and} \quad s = \frac{c}{d},$$

for some integers a, b, c , and d with $b \neq 0$ and $d \neq 0$. Furthermore, $c \neq 0$ because $c = sd$ and neither s nor d equals 0. By substitution and the laws of algebra,

$$\frac{r}{s} = \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}.$$

Now ad and bc are integers because a, b, c , and d are integers and products of integers are integers. Also $bc \neq 0$ by the zero product property because $b \neq 0$ and $c \neq 0$. Thus r/s can be written as a quotient of integers with a nonzero denominator, and so r/s is rational. $\square$

□

Ex. 4.2.31. Prove that if a real number c satisfies a polynomial equation of the form

$$r_3x^3 + r_2x^2 + r_1x + r_0 = 0,$$

where r_0, r_1, r_2 , and r_3 are rational numbers, then c satisfies an equation of the form

$$n_3x^3 + n_2x^2 + n_1x + n_0 = 0,$$

where n_0, n_1, n_2 , and n_3 are integers.

SOLUTION . Suppose c is a real number that is a root of a polynomial

$$p(x) = r_3x^3 + r_2x^2 + r_1x + r_0 = 0,$$

where n is a nonnegative integer, $r_3 \neq 0$, and r_0, r_1, r_2, r_3 are all rational numbers.

By definition of rational, there exist integers a_0, a_1, a_2, a_3 and b_0, b_1, b_2, b_3 such that $r_i = a_i/b_i$ and $b_i \neq 0$ for all integers i with $0 \leq i \leq 3$.

Since c is a root of $p(x)$,

$$r_3c^3 + r_2c^2 + r_1c + r_0 = 0.$$

By substitution,

$$\frac{a_3}{b_3} \cdot c^3 + \frac{a_2}{b_2} \cdot c^2 + \frac{a_1}{b_1} \cdot c + \frac{a_0}{b_0} = 0. \quad (*)$$

Multiplying both sides of $(*)$ by $b_0b_1b_2b_3$ gives

$$b_0b_1b_2a_3 \cdot c^3 + b_0b_1b_3a_2 \cdot c^2 + b_0b_2b_3a_1 \cdot c + b_1b_2b_3a_0 = 0.$$

which is a polynomial with integer coefficients. (Let $n_3 = b_0b_1b_2a_3$, $n_2 = b_0b_1b_3a_2$, $n_1 = b_0b_2b_3a_1$, and $n_0 = b_1b_2b_3a_0$. Then n_0, n_1, n_2 , and n_3 are integers since they are products of integers.) Thus c satisfies a polynomial with integer coefficients.

Remarks: The same proof may be generalized (by replacing “3” with “ n ”) to answer the next question 4.2.32 in the text book. The solution is provided below, if you are interested. $\square$

Definition: A number c is called a **root** of a polynomial $p(x)$ if and only if $p(c) = 0$.

Ex. 4.2.32. Prove that for all real numbers c , if c is a root of a polynomial with rational coefficients, then c is a root of a polynomial with integer coefficients.

SOLUTION . Suppose c is a real number that is a root of a polynomial

$$p(x) = r_n x^n + r_{n-1} x^{n-1} + \cdots + r_1 x + r_0 = 0,$$

where n is a nonnegative integer, $r_n \neq 0$, and $r_0, r_1, \dots, r_n$ are all rational numbers.

By definition of rational, there exist integers $a_0, a_1, \dots, a_n$ and $b_0, b_1, \dots, b_n$ such that $r_i = a_i/b_i$ and $b_i \neq 0$ for all integers i with $0 \leq i \leq n$.

Since c is a root of $p(x)$,

$$r_n c^n + r_{n-1} c^{n-1} + \cdots + r_1 c + r_0 = 0.$$

By substitution,

$$\frac{a_n}{b_n} \cdot c^n + \frac{a_{n-1}}{b_{n-1}} \cdot c^{n-1} + \cdots + \frac{a_1}{b_1} \cdot c + \frac{a_0}{b_0} = 0. \quad (**)$$

For each $i = 0, 1, 2, \dots, n$, let m_i be the product of a_i and all the b_j except b_i . In other words,

$$m_i = a_i b_0 b_1 \cdots b_{i-1} b_{i+1} \cdots b_{n-1} b_n.$$

Then each m_i is an integer (being a product of integers). Multiplying both sides of $(**)$ by $b_0 b_1 \cdots b_n$ gives

$$m_n c^n + m_{n-1} c^{n-1} + \cdots + m_1 c + m_0 = 0.$$

Hence c satisfies the equation

$$m_n c^n + m_{n-1} c^{n-1} + \cdots + m_1 c + m_0 = 0,$$

where $m_0, m_1, \dots, m_n$ are all integers. Thus c satisfies a polynomial with integer coefficients. □